

BARENTSBURG, Svalbard and Jan Mayen

WMO: 201070

Lat: **78.056N**Lon: **14.221E**Elev: **249**StdP: **14.56**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-10.5	-6.8	-17.4	2.2	-9.8	-13.5	2.7	-5.3	33.5	29.1	28.5	25.9	4.3	40	0.904

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.1	51.8	46.4	49.4	44.9	47.4	43.7	46.9	50.5	45.5	48.5	44.3	46.9	6.3	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
44.7	44.1	47.6	43.4	41.9	46.4	42.2	40.1	45.5	18.6	50.7	17.9	48.6	17.3	47.1	54.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.9	22.1	19.0	DB	-9.2	56.2	6.1	3.7	-13.6	58.9	-17.2	61.0	-20.7	63.1	-25.1	65.7	(4)
(5)			WB	-10.0	49.7	5.8	2.1	-14.1	51.2	-17.5	52.5	-20.8	53.7	-25.0	55.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	17.0	15.5	13.3	17.8	28.3	37.7	43.9	42.0	35.9	26.8	21.4	18.3
	DBStd	13.12	11.53	11.59	10.11	8.90	5.30	3.51	3.21	3.49	4.84	6.98	8.47	9.72
	HDD50	8555	1023	967	1139	965	671	368	191	249	422	718	858	984
	HDD65	14026	1488	1387	1604	1415	1136	818	653	713	872	1183	1308	1449
	CDD50	4	0	0	0	0	0	0	3	1	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.9	9.4	9.4	8.6	7.4	7.1	6.3	6.9	6.4	7.3	8.0	9.2	9.2
Precipitation	PrecAvg	16.7	1.8	1.4	1.5	1.0	0.8	0.6	1.1	1.3	1.6	1.8	1.9	1.9
	PrecMax	27.4	4.4	3.0	3.9	3.0	2.4	2.5	2.8	3.5	3.4	5.4	5.6	4.8
	PrecMin	9.9	0.1	0.0	0.2	0.0	0.3	0.1	0.2	0.4	0.4	0.2	0.7	0.4
	PrecStd	3.1	1.2	1.0	1.0	0.7	0.5	0.5	0.7	0.7	0.9	1.2	1.2	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.9	37.8	35.9	37.3	39.1	47.8	56.3	55.3	47.4	43.1	39.4	37.8
		MCWB	35.5	35.2	33.6	34.6	35.8	42.8	47.9	47.6	44.0	40.7	36.3	35.5
	2%	DB	36.2	35.9	33.8	35.2	37.4	45.9	53.0	50.8	45.5	41.2	37.0	35.0
		MCWB	34.2	34.1	31.7	33.3	34.7	42.0	47.2	45.7	42.7	38.7	34.3	32.5
	5%	DB	34.2	33.4	31.3	33.4	35.9	44.3	50.8	48.1	44.0	39.0	35.1	33.0
		MCWB	32.2	31.5	29.6	31.5	33.5	40.6	45.7	44.2	41.8	36.9	32.8	31.0
	10%	DB	32.0	30.8	27.9	31.0	34.7	42.9	48.8	46.4	42.4	36.7	32.9	31.2
		MCWB	30.0	28.8	26.3	29.1	32.5	39.6	44.6	43.3	40.4	34.7	30.7	29.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.1	35.6	34.2	35.5	36.8	44.0	50.0	48.8	45.0	41.4	37.0	35.9
		MCDB	37.4	37.0	35.3	36.4	38.0	46.9	54.0	53.7	46.5	42.7	39.1	37.5
	2%	WB	34.4	34.3	32.0	33.8	35.2	42.3	47.6	46.3	43.4	39.3	34.6	32.9
		MCDB	35.8	35.7	33.7	35.4	36.6	45.3	51.4	49.2	44.9	40.8	36.6	34.3
	5%	WB	32.4	31.6	29.5	31.6	34.0	41.1	46.3	44.9	42.2	36.8	33.0	31.3
		MCDB	33.9	33.0	31.1	33.3	35.5	44.0	49.9	47.4	43.7	38.8	34.9	32.7
	10%	WB	30.0	28.9	26.4	29.2	32.7	39.8	45.0	43.9	40.7	34.8	30.8	29.3
		MCDB	32.0	30.8	27.8	31.0	34.4	42.5	48.1	46.1	42.3	36.6	32.8	31.3
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.6	7.2	6.7	4.8	4.0	5.1	4.8	4.6	5.4	6.5	7.1
		MCDBR	7.5	8.6	9.6	6.6	5.1	5.8	8.2	6.9	6.0	6.6	7.2	8.7
		MCWBR	7.0	8.4	9.5	6.2	4.3	4.2	5.0	4.4	5.3	5.8	6.3	8.3
	5% WB	MCDBR	7.8	9.3	10.0	6.7	4.7	5.7	7.1	6.3	5.8	6.3	7.5	9.1
		MCWBR	7.4	9.3	10.1	6.4	4.4	4.4	4.7	4.4	5.4	5.9	6.9	9.1
Clear-Sky Solar Irradiance	taub	N/A	0.188	0.248	0.285	0.302	0.296	0.301	0.290	0.246	0.176	N/A	N/A	
	taud	N/A	1.918	2.101	2.245	2.288	2.402	2.468	2.463	2.374	1.966	N/A	N/A	
	Ebn at Noon	0	13	180	244	262	272	265	246	199	10	0	0	
	Edh at Noon	0	2	16	25	29	27	24	20	12	1	0	0	
All-Sky Solar Radiation	RadAvg	0	8	247	896	1500	1785	1463	928	349	42	0	0	
	RadStd	0	1	23	109	119	117	105	75	31	3	0	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+2.74	+8.33	+7.38	N/A	N/A	N/A	-994	-988	N/A	N/A

Nomenclature: See separate page